

```
        return result

print(fibonacci(99))
```

Compared to the additions required to implement our own cache in *fibonacciByRecursionMemoized.py*, using Python's `@lru_cache()` decorator is much simpler. Normally, calculating `fibonacci(99)` with the recursive algorithm would take a few centuries. With memoization, our program displays the 218922995834555169026 result in a few milliseconds.

Memoization is a useful technique for recursive functions with overlapping subproblems, but it can be applied to any pure function to speed up runtime at the expense of computer memory.